A Machine Learning Framework for Predicting Pulmonary Sarcoidosis Using Multi-Cohort Gene Expression

Report submitted to Meharry Medical College

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by

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under the guidance of

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Certificate of Capstone Project Completion

This is to certify that the report entitled **Predicting Pulmonary Sarcoidosis Using Machine Learning on Multi-Cohort Gene Expression Data**, submitted by **Jalen Jackson** to Meharry Medical College, is a record of bona fide research and practical work carried out under our supervision. The work presented in this report meets the academic and professional standards required for the successful completion of the Capstone Project as part of the Master of Science program at the Institute. In recognition of the student's dedication and scholarly contribution, this report is hereby deemed worthy of consideration for fulfillment of the degree requirements.

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- To my parents, Corey and Vickie Jackson, thank you for everything you have done for me. I would not be where I am today without your support, sacrifices, and constant belief in me. You have always pushed me to keep going, even when things got difficult, and for that I am forever grateful. I also want to thank my friends and family who supported me throughout this journey, always checking on me, encouraging me, and reminding me why I started. A special thank you to my undergraduate family at Texas Southern University for helping shape me into who I am today and for being a major part of my foundation. To everyone who played a role in helping me get to this point, I truly appreciate you all -

Declaration

I certify that:

a. the work contained in this report is original and was carried out by me under the guidance of my supervisor;

b. the work has not been submitted to any other institution for any other degree or diploma;

c. I followed the guidelines provided by the Institute in preparing this report;

d. I conformed to ethical norms and guidelines while conducting and presenting this work;

e. whenever Data were obtained, figures, methods, models, or text from other sources, I gave due credit by citing them appropriately in the text and references.

Jalen Jackson

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Finally, I acknowledge the researchers and public data repositories whose work made this project possible. Access to publicly available transcriptomic datasets provided the foundation for the analysis presented in this report.

Executive Summary

Sarcoidosis is a multisystem inflammatory disease characterized by granuloma formation and variable clinical presentation. Because its symptoms overlap with other pulmonary and inflammatory conditions, diagnosis remains challenging and often requires invasive procedures. The lack of reliable non-invasive biomarkers continues to limit early detection and clinical decision-making.

In this study, a machine learning framework was developed to classify pulmonary sarcoidosis using integrated multi-cohort gene expression data. Three publicly available GEO datasets (GSE18781, GSE42834, and GSE83456) were combined to improve generalizability. After preprocessing and harmonization, the merged dataset contained 398 samples (229 controls and 169 sarcoidosis cases) and 16,622 shared genes.

To address high dimensionality, a structured feature selection pipeline was applied. The feature space was reduced from approximately 39,000 genes to 16,622 shared genes, followed by further reduction using variance thresholding and SelectKBest, resulting in a final set of 1,000 features used for model training. Random Forest feature importance was also used to identify biologically relevant genes.

Multiple machine learning models were evaluated, including Logistic Regression, Random Forest, Support Vector Machine, K-Nearest Neighbors, Naive Bayes, and Gradient Boosting. A leave-one-cohort-out cross-validation strategy was used to assess generalizability. Random Forest achieved the best performance with an accuracy of 0.76 and ROC-AUC of 0.85.

These results demonstrate that combining multi-cohort integration with structured feature selection improves classification performance and model generalizability while providing

biological insight into sarcoidosis.

Keywords: Sarcoidosis, machine learning, gene expression, transcriptomics, multi-cohort integration, feature selection, Random Forest

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List of Abbreviations

AUC	Area Under the Curve
CS & DS	Computer Science and Data Science
GEO	Gene Expression Omnibus
KNN	K-Nearest Neighbors
LR	Logistic Regression
NB	Naive Bayes
PCA	Principal Component Analysis
RF	Random Forest
ROC	Receiver Operating Characteristic
SVM	Support Vector Machine

Chapter 1

Introduction

In this study, the focus was placed on predicting pulmonary sarcoidosis using machine learning on multi-cohort gene expression data. Three independent GEO datasets were combined to create a more realistic and challenging classification problem. This approach was designed to improve robustness across datasets and better reflect real-world biomedical variability.

1.1 Background

Pulmonary sarcoidosis remains difficult to diagnose because its presentation is highly variable. Some patients may have mild disease, while others may have progressive inflammatory involvement affecting multiple organ systems. Because symptoms and radiographic findings can overlap with other conditions, diagnosis often involves multiple steps and may still require invasive procedures such as biopsy. This creates a continued need for non-invasive strategies that can support clinical decision-making.

Gene expression analysis offers one possible path toward that goal. Because inflammatory disease states influence molecular activity, transcriptomic profiles can reflect biological processes associated with disease presence or progression. However, transcriptomic data are high-dimensional and often noisy. This makes machine learning especially appealing, since it can identify informative patterns across many features at once.

At the same time, machine learning on transcriptomic data comes with important challenges. These include high feature-to-sample ratios, platform heterogeneity, batch effects, and the risk of overfitting. As a result, careful preprocessing, feature selection, model comparison, and validation are necessary to produce reliable findings. These considerations shape the overall design of this capstone project.

Previous studies have shown that gene expression profiling can be useful for identifying disease-associated molecular patterns in inflammatory and immune-related conditions [9,11]. In sarcoidosis research, transcriptomic analyses have been used to investigate inflammatory signatures, immune activation, and potential biomarkers [9,11]. More recent studies continue to support the use of genomic and transcriptomic approaches for understanding complex diseases and identifying clinically relevant biomarkers [12,13]. These studies highlight that gene expression data can capture meaningful biological differences between disease and control groups.

However, a major limitation of many prior studies is the use of single-cohort datasets. Models developed and tested within a single dataset may perform well internally but often fail to perform consistently across independent datasets [10]. This limitation is particularly significant in genomics, where differences in experimental platforms, preprocessing methods, and cohort composition can strongly influence results.

Feature selection is another important focus in the literature. In high-dimensional datasets, reducing the number of features is often necessary to improve model performance, reduce overfitting, and enhance interpretability [8]. Common approaches include univariate statistical selection, regularization-based methods such as LASSO, and tree-based feature importance techniques.

In addition, machine learning methods such as Random Forest and Support Vector Machine have been widely applied in genomic classification tasks because of their ability to handle complex, nonlinear relationships [5,6]. More recent work has further demonstrated the effectiveness of machine learning approaches in healthcare and biomedical data analysis [14]. These approaches have shown strong performance in various biomedical applications, particularly when combined with effective preprocessing and feature selection strategies.

Despite these advances, many studies still lack external validation and rely on limited datasets. This reduces confidence in the general applicability of their findings.

This study builds on prior work by integrating multiple independent sarcoidosis cohorts, applying structured feature selection, and comparing several machine learning models within a unified workflow. By doing so, it addresses key limitations in previous studies, including limited robustness across datasets and limited interpretability of results.

1.2 Research Gap

Despite the progress made in applying gene expression analysis and machine learning to disease classification, several important limitations remain in the existing literature. Many prior studies rely on single-cohort datasets, which limits the ability to evaluate model performance across independent populations [10]. As a result, models that perform well within a single dataset may not maintain consistent performance when applied to new data.

In addition, validation across independent cohorts is often limited, reducing confidence in the robustness of reported results. Differences in experimental platforms, preprocessing methods, and cohort composition can further affect model performance, making cross-dataset consistency a key challenge in transcriptomic analysis [14].

Feature selection is another area that is frequently underutilized in high-dimensional genomic data [8]. Without proper feature reduction, models may suffer from overfitting and reduced interpretability. Furthermore, many studies focus primarily on prediction accuracy without providing meaningful biological interpretation of the selected features.

To address these limitations, this study integrates multiple independent sarcoidosis cohorts, applies structured feature selection methods, and evaluates model performance across datasets. This approach improves robustness across datasets while also identifying biologically relevant genes associated with disease processes.

Chapter 2

Objectives

The main objective of this study was to develop a machine learning framework for the classification of pulmonary sarcoidosis using integrated multi-cohort gene expression data.

The working hypothesis was that integrating multiple independent cohorts and applying structured feature selection would improve the performance and robustness of classification models.

2.1 Specific Objectives

- To develop a machine learning model for sarcoidosis classification.
- To improve generalizability by integrating multiple transcriptomic cohorts.
- To apply structured feature selection techniques to reduce dimensionality and remove noise.
- To identify biologically meaningful genes associated with sarcoidosis.
- To compare the performance of several machine learning algorithms.
- To evaluate classification performance using multiple quantitative metrics.

2.2 Significance

This study is significant because it addresses an important gap in sarcoidosis research. Many previous studies have relied on single-cohort datasets, which limits the ability to assess generalizability across different populations. By integrating multiple independent cohorts, a more robust and realistic analytical framework was developed, better reflecting the variability present in real-world biomedical data.

This study is also significant because it combines predictive modeling with interpretability. In addition to classification performance, feature selection and feature importance methods were used to identify genes associated with sarcoidosis. This provides biological insight into disease-related pathways and enhances the relevance of the findings from both a data science and biomedical perspective.

Furthermore, the structured feature selection approach used in this study addresses challenges associated with high-dimensional transcriptomic data, including noise, redundancy, and overfitting. By reducing the feature space from tens of thousands of genes to a smaller, more informative subset, the study improves both model performance and interpretability.

2.3 Organization of the Report

The remainder of this report is organized as follows. Chapter 3 presents the design and methodology, including dataset description, data integration, preprocessing, feature selection, model development, and validation strategy. Chapter 4 presents the results and discussion, including principal component analysis (PCA), model performance comparison, ROC analysis, confusion matrix evaluation, and interpretation of top gene features.

Chapter 5 presents the conclusion and outlines directions for future work. The appendix provides supplemental materials, including additional details on preprocessing, feature selection, and model interpretation.

Chapter 3

Design and Methodology

3.1 Proposed Method

The overall workflow followed a structured, step-by-step pipeline beginning with data collection from three publicly available GEO datasets. After identifying the relevant cohorts, gene identifiers were harmonized across studies, and only the shared genes present in all datasets were retained. Preprocessing and scaling steps were then applied to improve comparability across cohorts.

Following preprocessing, feature reduction and selection methods were applied to address the high dimensionality of gene expression data. This included the use of variance thresholding to remove low-variance genes and SelectKBest to retain the most informative features. In addition, Random Forest feature importance was used to identify key genes contributing to classification performance.

After constructing the feature matrix, Principal Component Analysis (PCA) was used to visualize the structure of the merged dataset. Multiple machine learning models were then trained and evaluated, and their performance was compared using quantitative metrics. Finally, the results were interpreted in terms of both predictive performance and biological relevance.

3.2 Tools and Technologies

The computational work was completed for this project in Python using a Google Colab environment. Core tools and libraries included:

- Python for data processing and modeling
- Pandas and NumPy for data handling
- Scikit-learn for machine learning, feature selection, preprocessing, and evaluation [4]
- Matplotlib for data visualization
- GEO datasets from the NCBI Gene Expression Omnibus

These tools were appropriate for the project because they supported large-scale data processing, reproducible analysis, and machine learning workflows commonly used in biomedical data science.

3.3 Data Collection

For this study, three independent transcriptomic datasets were used from the Gene Expression Omnibus:

- GSE18781 [1]
- GSE42834 [2]
- GSE83456 [3]

We selected these cohorts because they were relevant to sarcoidosis and contained gene expression data that could be aligned for integrated analysis. After preprocessing and harmonization, the merged dataset contained **398 total samples**, including **229 healthy controls** and **169 sarcoidosis samples**, with **16,622 shared genes** retained for modeling.

Table 3.1: Summary of merged dataset

Category	**Count**
Total Samples	398
Healthy Controls	229
Sarcoidosis Samples	169
Shared Genes	16,622
Number of Cohorts	3

3.4 Data Integration Challenges

Integrating multiple transcriptomic cohorts introduced several challenges that affected preprocessing and analysis:

- cross-platform heterogeneity between Affymetrix and Illumina platforms;
- probe ID to gene symbol mapping;
- batch effects across independent cohorts;
- alignment of shared gene features;
- ensuring comparable expression scales across datasets.

These issues were important because they could influence downstream model performance if not addressed carefully. A multi-cohort design improved realism and generalizability, but it also increased technical complexity.

3.5 Data Preprocessing and Feature Selection

The initial merged dataset contained approximately 39,000 gene features across all samples. Due to the high dimensionality of transcriptomic data, feature selection was necessary to

reduce noise, improve model performance, and prevent overfitting.

After harmonization and alignment of gene identifiers across the three GEO datasets, only shared genes were retained. This reduced the feature space to 16,622 genes.

A Variance Threshold method was applied to remove genes with very low variability across samples. This step reduced the number of features to 16,622 genes.

Following this, SelectKBest was applied to identify the most informative genes based on statistical significance. This reduced the feature set to 1,000 genes used for model training.

In addition, Random Forest feature importance was used to identify genes that contributed most strongly to classification performance. This method captured nonlinear relationships between features and disease outcomes.

Variance Threshold removes non-informative features based on variability, SelectKBest selects statistically significant features, and Random Forest identifies biologically meaningful features. The combination of these methods provided effective dimensionality reduction while maintaining interpretability. The variance threshold step did not reduce the number of features, indicating that most genes retained sufficient variability across samples.

Overall, the feature selection process reduced dimensionality from approximately 39,000 genes to 1,000 features, improving efficiency, reducing overfitting, and enhancing model performance.

Table 3.2: Feature selection and dimensionality reduction process

Step	**Number of Features**
Initial dataset	∼39,000
After gene matching	16,622
After variance threshold	16,622
After SelectKBest	1,000

3.6 Data Analysis

Principal Component Analysis (PCA) was used for dimensionality reduction and visualization. PCA enabled the projection of the merged dataset into a lower-dimensional space, allowing patterns of separation between sarcoidosis and control samples to be examined.

3.7 ML Classifier

For supervised classification, six machine learning models were trained and evaluated:

- Logistic Regression
- Random Forest
- Support Vector Machine
- K-Nearest Neighbors
- Naive Bayes
- Gradient Boosting

These models were selected to compare both linear and nonlinear approaches. Logistic Regression served as a baseline model. Random Forest and Gradient Boosting were included because tree-based methods perform well on structured and nonlinear data. Support Vector Machine was included due to its effectiveness in high-dimensional classification tasks. K-Nearest Neighbors and Naive Bayes were included as additional benchmark models for comparison.

3.8 Validation Strategy

A leave-one-cohort-out cross-validation strategy was used to evaluate model generalizability across independent datasets. Model performance was assessed using accuracy and ROC-

AUC, along with additional evaluation metrics including precision, recall, F1-score, and confusion matrix analysis for the best-performing model.

This validation approach was selected to reduce overfitting and improve the reliability of performance estimates. In high-dimensional biomedical datasets, proper validation is critical because apparent performance can be inflated when models capture noise rather than true disease-related patterns. By evaluating performance across independent cohorts, the robustness and generalizability of the models were more effectively assessed.

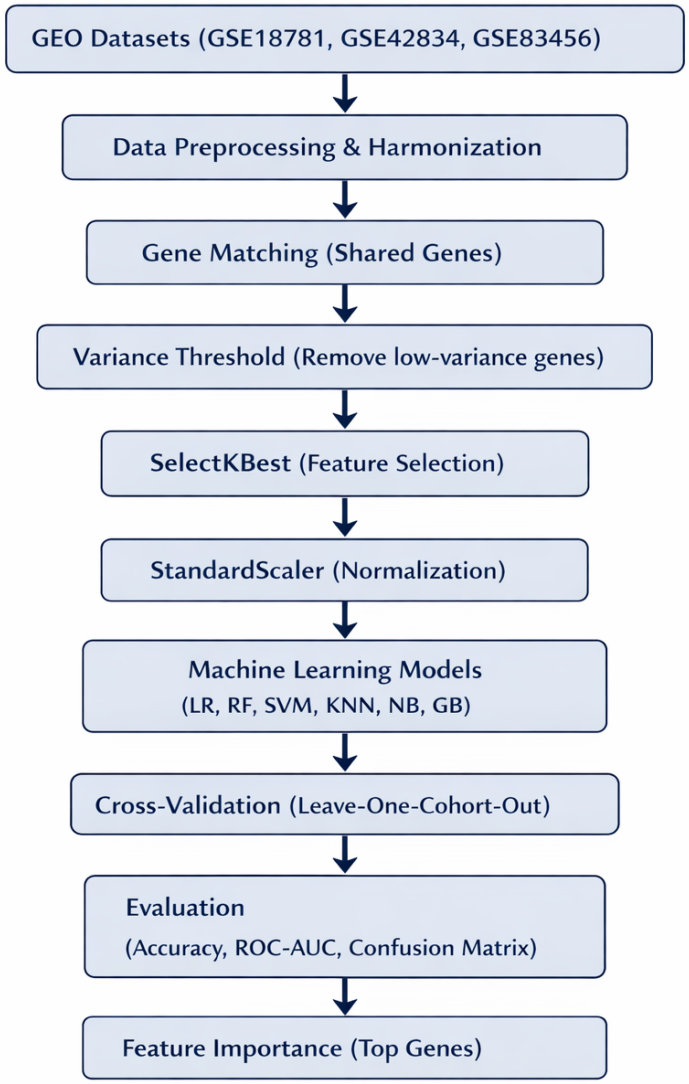

Figure 3.1: Overall workflow of data integration, preprocessing, feature selection, model training, and evaluation.

Chapter 4

Results and Discussion

4.1 Principal Component Analysis

Principal Component Analysis (PCA) was used to visualize the structure of the merged dataset. PCA reduces the dimensionality of the gene expression data while preserving the most important sources of variation, allowing underlying patterns to be observed.

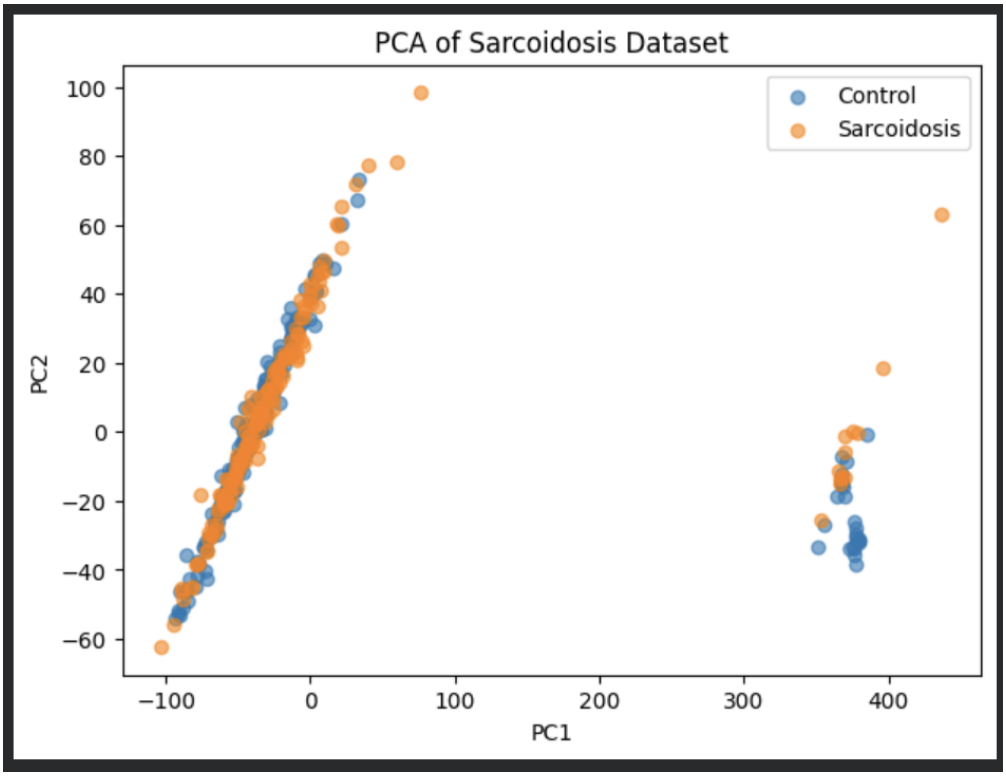

Figure 4.1: PCA of the merged dataset showing partial separation between control and sarcoidosis samples.

The PCA plot showed partial separation between sarcoidosis and control samples, indicating that the merged data retained meaningful disease-related structure. However, some overlap remained, reflecting both the biological complexity of sarcoidosis and the inherent variability present in multi-cohort transcriptomic data. This suggests that while disease-associated patterns exist, the classification problem is non-trivial and requires robust modeling approaches.

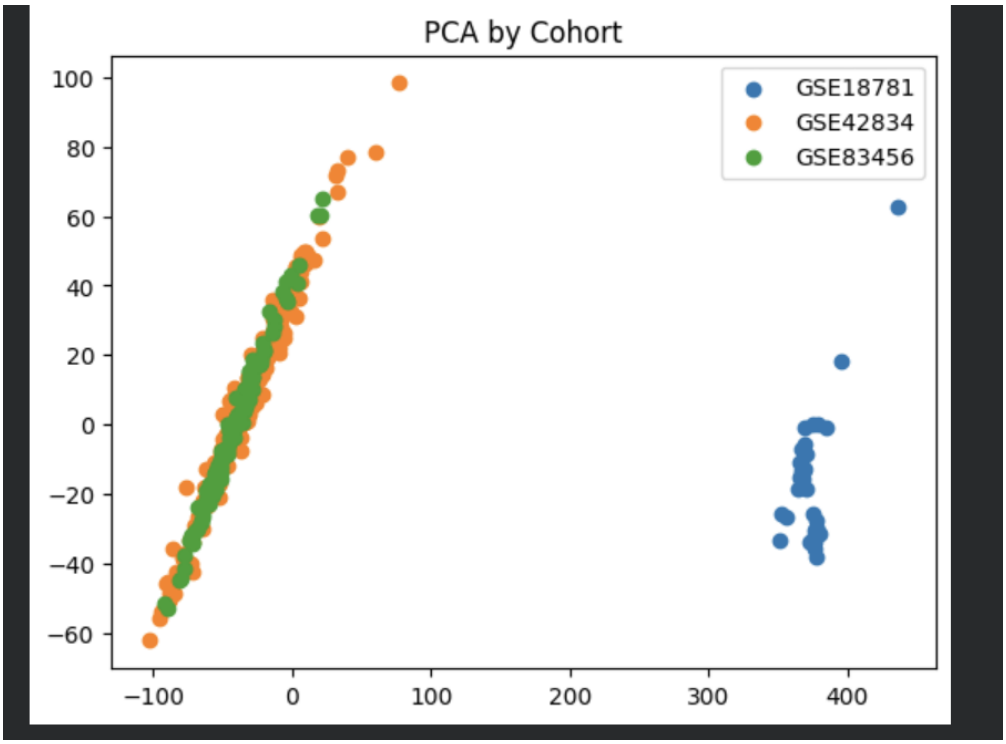

Figure 4.2: PCA of the merged dataset colored by cohort, showing clustering patterns associated with different datasets.

When visualized by cohort, PCA revealed clear clustering patterns corresponding to individual datasets. Samples from different cohorts formed distinct groups, indicating the presence of batch effects and platform-specific variation. This highlights a key challenge in multi-cohort gene expression analysis, where technical differences across studies can influence data structure.

These findings reinforce the importance of a multi-cohort validation strategy. Models trained on a single dataset may capture cohort-specific patterns rather than true disease-related signals. By using leave-one-cohort-out cross-validation, this study ensures that model performance reflects generalizable biological patterns rather than dataset-specific bias.

4.2 Model Performance Comparison

The machine learning models exhibited varying performance in classifying sarcoidosis from healthy controls. Random Forest achieved the strongest overall performance, with an accu-

racy of 0.761307 and an ROC-AUC of 0.850366. Support Vector Machine also demonstrated strong performance, while Logistic Regression produced moderate results. Gradient Boosting and K-Nearest Neighbors showed comparatively lower performance, and Naive Bayes performed the weakest among the evaluated models.

Table 4.1: Model performance comparison

Model	**Accuracy**	**ROC-AUC**	**Precision**	**Recall**	**F1-Score**
Random Forest	0.7613	0.8504	0.7256	0.7041	0.7147
SVM	0.7362	0.8484	0.6404	0.8639	0.7355
Logistic Regression	0.7161	0.7990	0.6186	0.8639	0.7210
Gradient Boosting	0.6608	0.7331	0.5752	0.7692	0.6582
KNN	0.6231	0.6746	0.5446	0.6864	0.6073
Naive Bayes	0.4246	0.5000	0.4246	1.0000	0.5961

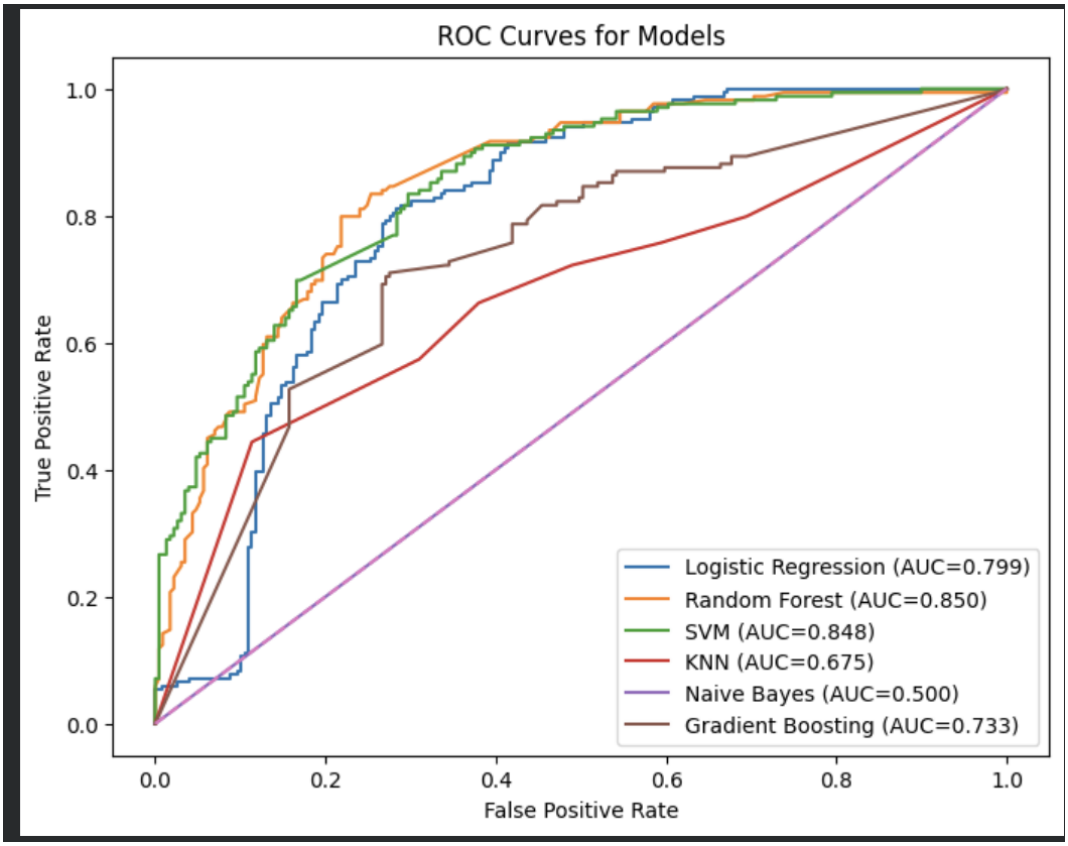

Figure 4.3: ROC curves comparing performance across all classification models. Random Forest and SVM show the highest performance.

These results indicate that nonlinear models are better suited for this classification task. Random Forest and SVM likely performed better because they can capture complex interactions among gene features, whereas linear models, such as Logistic Regression, are limited by their assumptions of linear separability.

Compared to previous studies that relied on single-cohort datasets, this study demonstrates improved consistency across independent datasets. The use of leave-one-cohort-out validation provides a more realistic assessment of model performance.

4.3 ROC Curve Analysis

The Receiver Operating Characteristic (ROC) curve was used to assess the discriminative ability of the classification model. The ROC curve was used to assess the discriminative

ability of the classification model. The AUC values ranged from approximately 0.82 to 0.85 across evaluation settings, indicating strong classification performance and demonstrating that the model performs substantially better than random guessing.

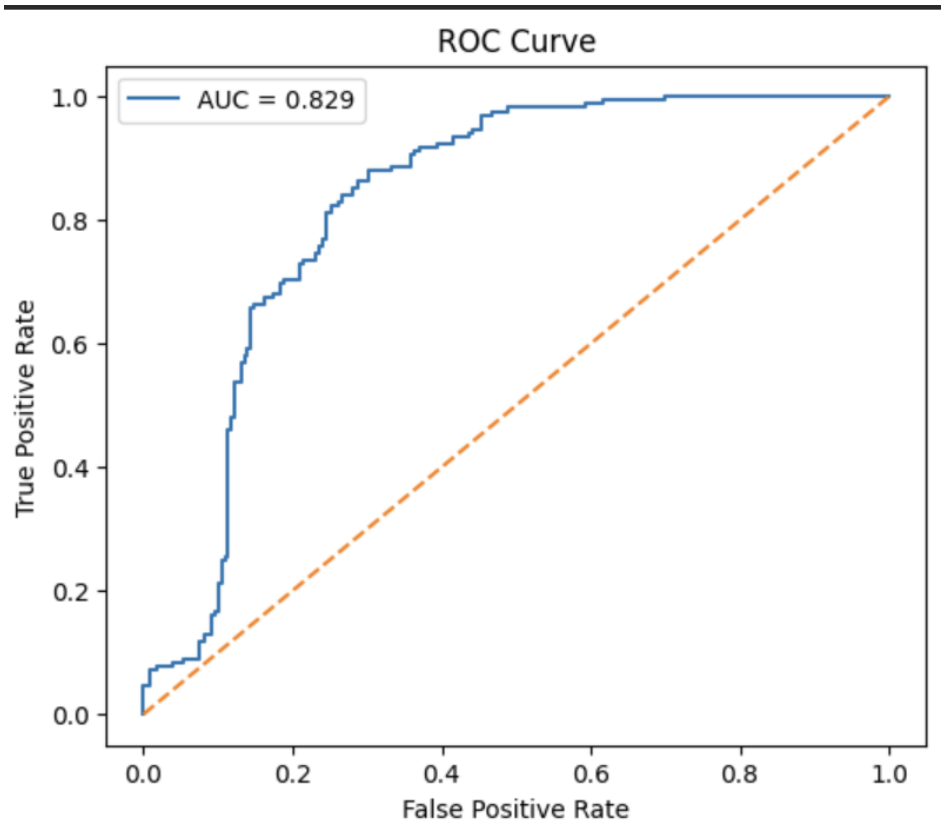

Figure 4.4: ROC curve of the best-performing model showing strong classification performance (AUC $\approx$ 0.83).

These results support the effectiveness of the overall workflow, including preprocessing, feature selection, and model development.

4.4 Confusion Matrix Analysis

A confusion matrix was used to provide a detailed breakdown of classification performance by displaying true positives, true negatives, false positives, and false negatives.

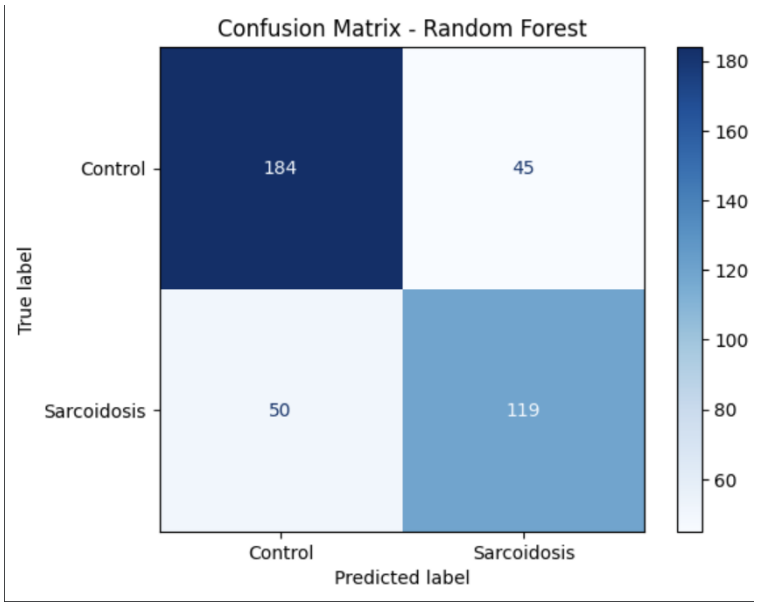

Figure 4.5: Confusion matrix for the Random Forest model using leave-one-cohort-out validation.

The confusion matrix shows that the Random Forest model correctly classified most sarcoidosis and control samples, with relatively few misclassifications. This indicates balanced performance across both classes and supports the reliability of the classification results.

4.5 Top Gene Features

Feature importance analysis identified several genes associated with immune and inflammatory pathways, including **CXCL10**, **FCGR1A**, **GBP5**, and **LEF1**. These genes provide biological context for the classification results and support the interpretability of the model.

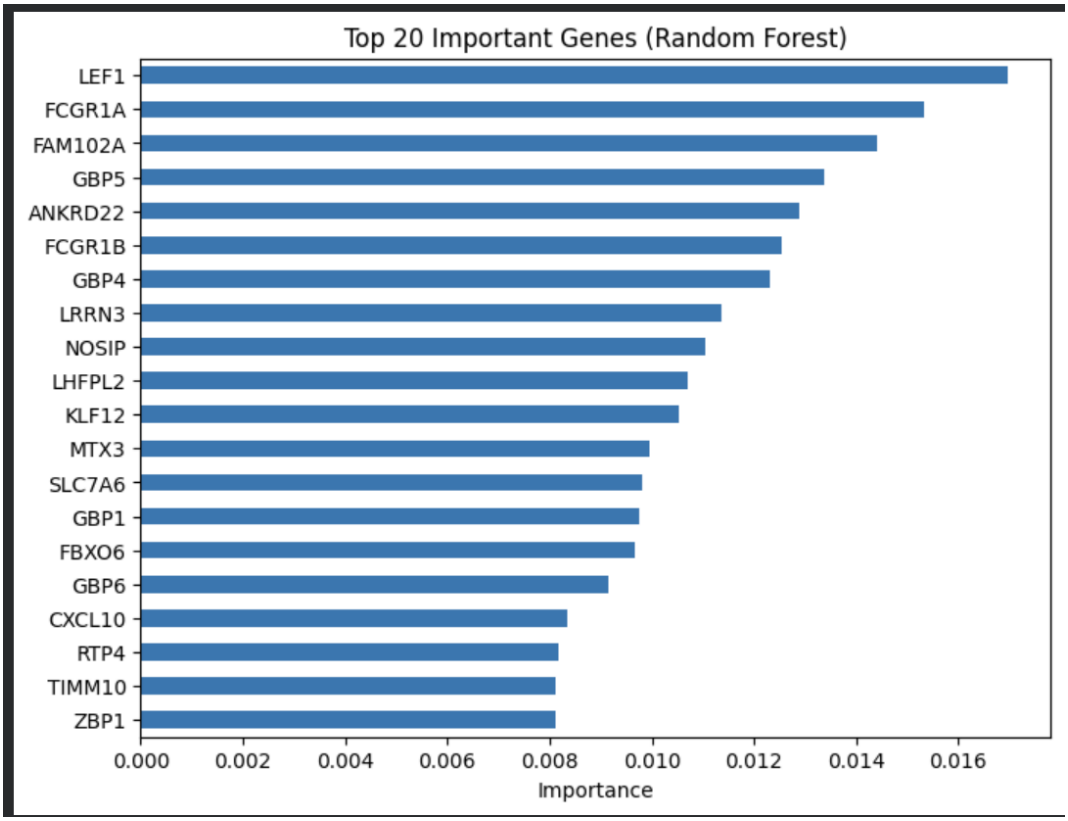

Figure 4.6: Top 20 most important genes identified using Random Forest feature importance.

The ability to identify biologically meaningful genes is important because it extends the value of the model beyond prediction and supports its relevance in biomedical research.

4.6 Discussion

The results of this study highlight the value of using multi-cohort data when developing machine learning models for biomedical classification tasks. PCA analysis revealed both disease-related structure and strong cohort-specific clustering, emphasizing the importance of robust validation strategies.

Random Forest demonstrated the strongest performance, likely due to its ability to handle high-dimensional data and capture nonlinear interactions among gene features. Support Vector Machine also performed well, further supporting the importance of nonlinear modeling approaches.

Feature selection played a critical role in improving both model performance and interpretability. In high-dimensional biomedical datasets, where the number of features far exceeds the number of samples, structured feature selection is essential to reduce noise and prevent overfitting.

Despite these strengths, several limitations remain. Cross-platform heterogeneity between datasets could not be fully eliminated, and batch effects may still influence the results. Additionally, the lack of detailed clinical metadata limited the ability to explore demographic factors.

Overall, these findings demonstrate that combining multi-cohort transcriptomic data with structured feature selection and machine learning provides a robust framework for sarcoidosis classification.

This study demonstrates that integrating multi-cohort transcriptomic data with robust validation strategies improves model robustness across datasets and provides a more realistic evaluation of performance compared to single-cohort approaches.

Additionally, the absence of complete demographic metadata, such as age and sex, limited the ability to perform subgroup analyses.

Chapter 5

Conclusion and Future Work

In this capstone project, a machine learning framework was developed to classify pulmonary sarcoidosis using integrated multi-cohort gene expression data. By combining three independent datasets from the Gene Expression Omnibus (GEO), this study addressed a key limitation in prior research, where models are often trained and evaluated on a single dataset. The use of a multi-cohort design provided a more realistic and challenging evaluation setting, better reflecting variability across populations and experimental conditions.

The results demonstrated that Random Forest achieved the strongest overall performance, with an accuracy of approximately 0.76 and an ROC-AUC of approximately 0.85. Support Vector Machine also performed well, further indicating that nonlinear models are effective for capturing complex transcriptomic patterns. PCA analysis revealed partial separation between disease and control samples, as well as clear cohort-specific clustering, highlighting both the presence of biological signal and the impact of batch effects.

Feature selection played an important role in improving model performance and interpretability. Reducing the dataset from approximately 39,000 genes to 1,000 informative features allowed the models to focus on the most relevant signals while minimizing noise. The identification of genes associated with immune and inflammatory pathways further supported the biological relevance of the model outputs.

Despite these strengths, several limitations remain. Cross-platform heterogeneity across

datasets could not be fully eliminated, and batch effects may still influence model performance. Additionally, the absence of complete demographic metadata, such as age and sex, limited the ability to perform subgroup analyses and assess potential biases across populations.

Future work will focus on validating the proposed workflow using additional independent datasets and incorporating more complete clinical metadata where available. Further improvements may include exploring advanced feature selection techniques such as LASSO and Elastic Net, as well as evaluating model performance using smaller feature subsets (e.g., 500 or 100 genes). Additional research may also investigate deep learning approaches and pathway-level analysis to enhance biological interpretation and improve predictive performance.

Overall, this study demonstrates that integrating multi-cohort transcriptomic data with structured feature selection and machine learning provides a robust and generalizable framework for sarcoidosis classification, contributing to the growing role of data science in biomedical research.

As I continue my work in clinical research, I plan to further develop this project by pursuing a Ph.D. at Meharry Medical College and expanding this research into more clinically applicable models.

Appendix A

Supplemental Materials

This appendix provides supporting materials related to reproducibility and transparency of the analysis presented in this study.

Code Availability

The code used in this project is publicly available on GitHub:

`https://github.com/2020jalenjackson-coder/capstone`

The repository contains separate notebooks for different stages of the analysis, allowing the workflow to be easily followed and reproduced.

Repository Structure

The GitHub repository is organized as follows:

- `Preprocess.ipynb` – Data cleaning, gene matching across cohorts, and feature selection
- `Final.ipynb` – Model training, evaluation, ROC analysis, and confusion matrix generation

- `Final Capstone Presentation (2).pptx` – Presentation slides summarizing the project

Reproducibility

All analyses were conducted using Python in a Google Colab environment using standard scientific computing libraries, including `pandas`, `NumPy`, and `scikit-learn`.

The workflow includes data preprocessing, multi-cohort integration, feature selection, model training, validation, and evaluation. The provided code enables replication of the results and supports further extension of this work.

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